

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Thursday, 20 August 2026, 9:09 AM**State** Finished**Completed on** Thursday, 20 August 2026, 9:18 AM**Time taken** 9 mins 13 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | int main()
3 | {
4 |     int count=0,num;
5 |     scanf("%d",&num);
6 |     for (int i = 1; i <= num; ++i)
7 |     {
8 |
9 |         if (num % i == 0)
10 |        {
11 |            count++;
12 |        }
13 |        count++;
14 |        count++;
15 |    }
16 |    count++;
17 |    printf("%d",count);
18 |
19 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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